



BEFORE

Since the end of the last ice age, natural factors and human actions have had an impact on Earth's climate.

NATURAL CLIMATE CHANGES

The last ice age **22–23** ended between 10,000 and 15,000 years ago, and since then the world has been getting warmer. Researchers have suggested a range of natural causes for this warming, including variations in Earth's orbit, fluctuations in the Sun's output, and volcanic activity.



CONGESTED ROADS

burning of fossil fuels, such as oil and coal, has led to **higher levels of carbon dioxide (CO₂)** in Earth's atmosphere, and so increased the "greenhouse effect."

THE GREENHOUSE EFFECT

In 1824, French mathematician and physicist **Joseph Fourier** observed that Earth's surface and atmosphere are warmed because the **Sun's heat is absorbed by naturally occurring "greenhouse" gases**, including methane (CH₄) and carbon dioxide (CO₂).

INDUSTRIAL EFFECTS

Since the Industrial Revolution **292–93**, the average global temperature has risen more quickly. Increased

Since the Industrial Revolution (see pp.292–93), average global temperatures have risen by about 1.4°F (0.8°C). This warming has accelerated in the last four decades. From 1996 to 2009, 13 of the warmest years on record have been noted.

The current situation

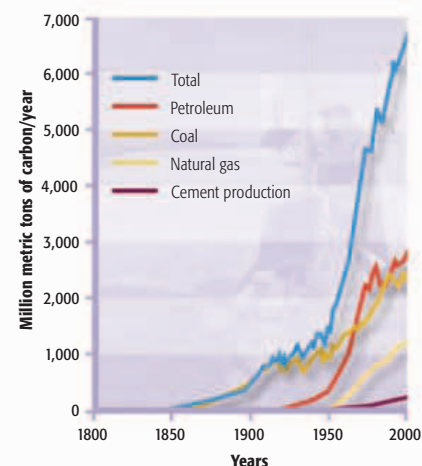
In 2007, the Intergovernmental Panel on Climate Change (IPCC) published a report in which it projected probable rises in temperature of between 3.2 and 7°F (1.8–4°C) by the end of the 21st century. The panel declared themselves confident that the "net

84 PERCENT of Antarctic glaciers have retreated since 1950.
20 PERCENT reduction in polar ice caps since 1978.
10 PERCENT shrinkage in global snow and ice cover since 1960.

effect of human activity since 1750 has been one of warming." The IPCC was set up by the World Meteorological Organization (WMO) and the United Nations Environment Program (UNEP) to investigate climate change. Its conclusion that humans have caused climate change is based on the fact that the atmospheric concentration of carbon dioxide (CO₂) has increased by 31 percent since the preindustrial era, intensifying the greenhouse effect (see BEFORE).

Increased CO₂ emissions are a direct result of industrialization. Coal-burning power stations generate CO₂—as do air, sea, and road traffic; each of the 232 million cars in the US produces more than five tons of CO₂ every year.

Plants and trees naturally counteract emissions, by incorporating CO₂ into their tissues, and by releasing oxygen into the atmosphere, but 13 million hectares (32 million acres) of the



Fossil carbon emissions

The graph indicates the rise in carbon emissions in the atmosphere since 1850. Steady increases in industrial activity and in vehicle numbers, along with growing CO₂ emissions caused by burning fossil fuels and producing cement and natural gas, mean that global fossil carbon emissions are now more than 7 billion tons per year.

Climate Change

Earth's atmosphere has a natural capacity to warm the planet. Without this heat, the planet could not sustain life—but the world is rapidly getting warmer. Many experts believe human activity is to blame, calling for urgent action to prevent a global crisis and protect the planet for future generations.

